CS2060 Programming with C - Ch. 6 Assignment

1 Instructions

1. Submit this assignment on Canvas.
2. Solve each question in a separate (.c) file.
3. Print screen of the execution of each question (as evidence that your code is working).
4. It is highly recommended to run your code (for each question) on the test cases and print screen the results each time.
5. For each question, collect the .c file and its print screens into a folder.
6. Create a folder for the whole assignment.
7. Place all question folders into the assignment folder.
8. Compress the assignment folder into one zipped folder.
9. Upload this zipped folder to Canvas.
10. It is your own responsibility to check that you uploaded the assignment correctly (you can open the submission web page on Canvas and download your submission to check everything is OK).
11. If the Canvas is down and the deadline is too close, you can send the assignment via email to me and the TA with a print screen of the Canvas page that is down (as evidence that the Canvas website is down).

2 Questions

1. Write a C program to perform the following:
    - Asking the user to enter using the keyboard the number of elements of the one dimensional array (positive integer number).

- Initializing its elements by random numbers ranging from 1 (included) to 10 (included).
- Calculating the Mean, Median, and Mode.
- The **mode** Function
  - It can handle a tie for the mode value.
  - **IF** there are 2 values with the same frequency, **THEN** the data is **bimodal** and both values should be displayed.
  - **IF** there are more than 2 values with the same frequency, **THEN** the data is **multimodal** and all the values with the same frequency should be displayed.
- The **median** Function
  - It calculates the average of the 2 middle elements in an array with an even number.

```
Please, enter the size of the array: -1
Invalid Input
```

```
Please, enter the size of the array: 0
Invalid Input
```

```
Please, enter the size of the array: 10
The mean is: 4.6000

The unsorted array is:
  1  3  9 10  2  4  7  2  7  1
The sorted array is:
  1  1  2  2  3  4  7  7  9 10
The median is: 3.5000

The data is multimodal.
The values are: 1 2 7
```

```
Please, enter the size of the array: 11
The mean is: 6.3636

The unsorted array is:
  9  8  6  3  7  3  7  8  8  7  4
The sorted array is:
  3  3  4  6  7  7  7  8  8  8  9
The median is: 7

The data is bimodal.
The values are: 7 8
```

```
Please, enter the size of the array: 100
The mean is: 5.0100

The unsorted array is:
  2  7  8  2  1  1 10  1  4  2  9  9  9  5  3  9  6  2  4  3
  4  1  2  6  8  6  2  6  1  8  4  2  6  3  4  9  4  3  9  7
  6  9  5  7  3  7  5  8 10 10  3  4  2  6  1  1  1  3  7  1
  2  2  2  7  5  7  5  8  1  5  6  9  5 10  7 10  8  1  9  9
  2  1  4  3  6  7  3  6  1  1  8  2  5  2 10  1 10  7 10  3
The sorted array is:
  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  2  2  2  2  2
  2  2  2  2  2  2  2  2  2  3  3  3  3  3  3  3  3  3  3  4
  4  4  4  4  4  4  4  5  5  5  5  5  5  5  5  6  6  6  6  6
  6  6  6  6  6  7  7  7  7  7  7  7  7  7  7  8  8  8  8  8
  8  8  9  9  9  9  9  9  9  9  9  9 10 10 10 10 10 10 10 10
The median is: 5.0000

The data is unimodal.
The value is: 1
```

2. Write a C program to perform the following:

   - Defining the following 2 integer arrays:

     ```
     int a[] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10};
     int b[] = {-1, 2, -5, 9, -10};
     ```

   - You have to define a recursive function called printArrayReverse that takes an array and the size of the array as arguments, prints the array in the reverse order, and returns nothing. The function should stop processing and return when it receives an array of size zero.

```
Array elements (in reverse order) are: 10  9   8   7   6   5   4
      3   2   1
Array elements (in reverse order) are: -10 9  -5   2  -1
```